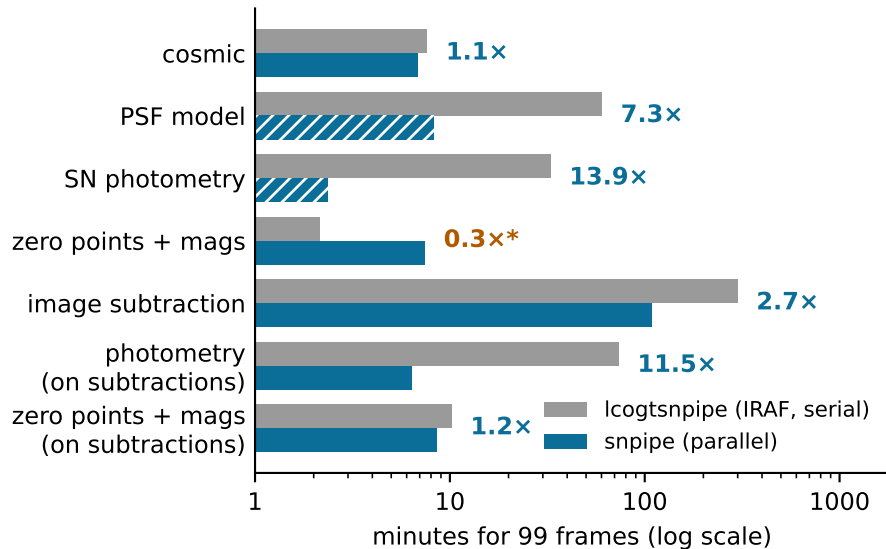
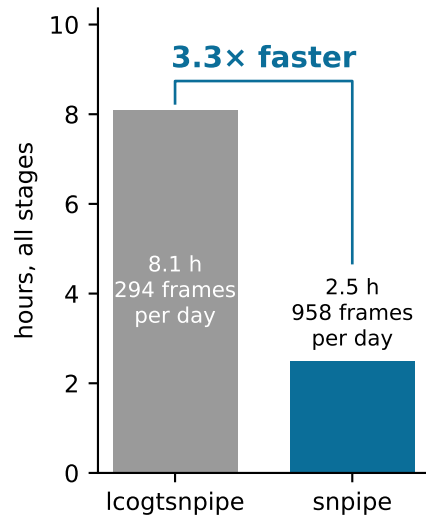


(a) per stage



(b) end to end



Same 99 LCO frames of SN 2024pxl, same 8-core / 16 GB node. Hatched: new time estimated from per-frame times. *Calibration was timed while two subtractions shared the node, plus ~27 s Python start-up per command; alone it takes 0.6-0.9 s per frame (old: 0.7 s).